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"SOLANGE": Quantum Computing and AI Against Non-Druggable Mutations - A New Frontier in NSCLC Oncology

SOLANGE is a CGU–KGI quantum-classical/AI research initiative targeting TP53, STK11, and KEAP1 mutations in non-small cell lung cancer that currently lack approved targeted therapies. Led by CGU Professor Itamar Shabtai with KGI Professor Animesh Ray and CGU doctoral researcher Doron Cohen together with Professor Molly Schmidt, the project integrates quantum computing, high-performance computing, AI, and governed provenance to determine when quantum simulation adds value beyond classical methods. Building on a functioning platform already validated on IBM quantum hardware, the BLAIS project will execute full-scale simulations, train CGU and KGI students, generate joint publications, and develop competitive external funding proposals.